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XAI4RE – Using Explainable AI for Responsible and Ethical AI

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Abstract

As Artificial Intelligence (AI) systems are increasingly integrated into high-stakes domains, the demand for transparency has become paramount. The opacity of “black-box” models poses significant challenges in trust, fairness, and accountability. Explainable AI (XAI) is a vital approach for addressing these concerns by enabling transparency, fostering trust, and ensuring ethical deployment across various sectors, including healthcare, human resources, finance, autonomous systems, and more. This paper explores how XAI methods can be used throughout the AI lifecycle for creating human-centered, ethical, and responsible AI systems by enhancing transparency, reducing bias, and protecting data privacy. Furthermore, the paper introduces XAI4RE, a theoretical framework that links XAI principles and purposes to concrete stages of the AI lifecycle, demonstrating how to address ethical considerations effectively. This approach involves engaging different stakeholders, such as developers, regulators, and users, at each stage. The framework highlights the critical role of XAI in promoting fairness, accountability, and human-centric design using general guidelines that discuss the relevant insights that can be drawn from XAI at each lifecycle stage. Ultimately, this paper underscores the importance of XAI in bridging the gap between technical advancements and ethical AI practices to foster societal trust and responsible systems.

CCS Concepts

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **HCI theory, concepts and models**; • **HCI design and evaluation methods**; • **Applied computing**; • **Law, social and behavioral sciences**; • **Computing methodologies**; • **Artificial intelligence**; • **Philosophical/theoretical foundations of artificial intelligence**;

Keywords

Artificial Intelligence, Explainability, XAI, Ethical and Responsible AI, Decision Making Systems, Users’ Trust, Fairness, Accountability, Human-Centered AI

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1 Introduction

As Artificial Intelligence (AI) systems grow more complex, particularly with the increasing reliance on opaque models such as Deep Neural Networks (DNNs), the demand for transparency in decision-making processes becomes more critical than ever [8, 45]. These “black-box” models, while powerful, often operate in ways that are hard to understand, which may lead to transparency, trust, and accountability challenges, especially for laypeople [19, 27, 29, 40]. Even a well-designed model risks generating inaccurate, unfair, or harmful outcomes, which may undermine users’ trust and the adoption of the system [28, 39]. Hence, understanding how the system works and/or why a specific decision was made is crucial, especially in domains that have a significant impact on our lives. Therefore, AI systems must go beyond mere functionality and provide actionable insights into their internal mechanisms [1, 16, 17, 39]. Moreover, various stakeholders, including researchers, developers, regulators, decision-makers, and end-users, are calling for creating clear explanations to justify AI decisions, ensuring that systems can be trusted and ethically deployed [36].

The main aim of explainable AI (XAI) is to bridge the gap between the system’s performance and the need for human comprehension [1, 26, 39]. XAI tools and methods make AI systems more transparent and interpretable by combining advanced computational techniques with insights from social sciences and psychology [14]. This balance between explainability and predictive accuracy is essential for fostering trust, improving collaboration between humans and AI, and enabling effective oversight and management of AI systems [14, 26]. Furthermore, XAI facilitates the alignment of AI behavior with human values by ensuring that its decisions can be scrutinized and refined [13]. Hence, XAI may promote accountability, enhance user engagement, and pave the way for responsible and ethical AI in a wide range of domains [1, 28, 39, 45].

This paper explores how different stakeholders can use XAI methods to align with the ethical considerations in the AI lifecycle. In addition, the paper proposes XAI4RE, a theoretical framework for using XAI for creating Responsible and Ethical AI systems. The framework suggests general guidelines for using XAI at the various stages of the AI lifecycle, concerning the diverse stakeholders, to ensure that the ethical considerations are addressed. Following

the suggested theoretical framework may help improve the transparency and fairness of AI systems while fostering user trust and creating responsible and ethical AI systems.

2 Related Work

AI is used across a wide range of domains, including business, healthcare, government, and beyond. By leveraging technologies such as machine learning and natural language processing, AI systems enhance the accuracy and efficiency of decision-making through automation, augmentation of human abilities, and the generation of data-driven insights [2].

Lately, various strategies for combining human and AI in the decision-making processes have been suggested [1, 26, 36, 41]. These strategies aim to balance the efficiency and consistency of AI with the contextual understanding and ethical oversight of human decision-makers. For example, McDermid et al. [26] argued that XAI methods often provide approximations and usually present partial information. Users' need for explanations is broadly ethical in nature and relates to whether the model's predictions and/or behavior align with people's expectations of fairness, respect, as well as the normative goals for the system. Additionally, Osasona et al. [36] argued that there is a need to ensure that AI decisions are consistent with society's values and expectations and that respecting individuals' rights to their data and obtaining clear consent are essential for deploying ethical AI.

2.1 Explainable AI (XAI)

XAI plays a crucial role in enhancing transparency and trust in AI decision-making [1, 28, 39, 45]. XAI types are varied and include textual/visual explanations, local/global explanations, explanations by examples/ simplification/ feature relevance/ specification, as well as a combination of the various types [1, 18]. XAI techniques, such as LIME, SHAP, and counterfactual explanations, aim to make complex AI models interpretable, addressing concerns about the opacity of black-box systems [28, 39]. Hence, using different XAI methods can increase the understanding of the system and thus allow the identification and correction of biases so that only significant and relevant variables will affect the decision of the system [1, 25, 26, 41, 48].

XAI facilitates human-AI collaboration by enabling decision-makers to justify AI-influenced choices [13]. However, despite its benefits, XAI challenges remain in balancing interpretability with model performance [39, 43], unintended negative consequences for various stakeholders [30], and the potential to mislead users to trust the system [9, 20]. Current AI regulations in the EU and USA emphasize explainability but argue that some technical limitations of XAI methods may lead to misleading or incomplete explanations. For example, the EU's proposed AI Act [15] introduces detailed transparency requirements that XAI could address. However, they emphasize that there are significant differences between XAI approaches and the regulatory requirements, and that collaboration between legal and technical experts is essential.

2.2 Stakeholders Needs from XAI

Various stakeholders have diverse needs and preferences for explanations [6, 36, 42], and XAI approaches should be tailored to specific

use cases, decision-makers, and users [1, 3, 10, 41, 47]. According to [1, 21, 22, 26], the main stakeholder groups are:

- **Developers:** A person or an organization that is involved in AI development, evaluation, and testing processes.
- **Regulators:** A person or an organization that is involved in the design, testing, and auditing process through legislative or standard-setting activities.
- **Users:** A person or a group that uses the system. The user can be an expert user (a decision maker who operates the system or has solid domain knowledge) or an end user (affected by the system output). The decision maker and the end user may be the same or different people.

Implementing XAI in practice requires a contextualized approach that considers stakeholder needs [1]. For example, expert users such as developers, regulators, and decision-makers often prefer minimal explanations, while others desire more comprehensive information [36]. Besides having different needs, there are also different benefits for the different stakeholders. Regulators can benefit from XAI by verifying legal compliance and reviewing algorithmic decisions [36]. Developers can use XAI to enhance model interpretability [25], improve model performance [48], improve fairness and understanding [42], and perform scientific evaluations and regulatory decision-making [1]. For end-users and decision-makers, XAI can improve their understanding, trust, and task performance [39, 41] and facilitate human-AI collaboration in the decision-making process [12, 13, 36].

2.3 Ethical Considerations

According to [11, 16, 21], three kinds of ethical frameworks can be considered: (1) Consequentialist Ethics: choosing the option with the best overall consequences after evaluating possible outcomes; (2) Virtue Ethics: consistent displaying of virtues like justice (i.e. advocating fairness and non-discrimination while promoting equality, diversity, and bias mitigation), honesty (i.e. promote organizational and technical transparency, while fostering accountability), responsibility (i.e. fostering accountability for AI technologies and their outcomes and mitigating the risk of unethical behavior) and care (i.e. understanding and addressing users' needs with empathy and motivating professionals to prevent harm, while promoting safety, sustainability, solidarity, and social cohesion), while reflecting moral values and earning social approval; and (3) Deontological Ethics: acting according to various duties and obligations, while adhering established social norms and ethical virtues.

Ethical considerations in AI systems include the understanding and use of justice theories, mitigation of biases, transparency, accountability, and implementation of ethical frameworks [19, 27, 29, 32, 35]. However, while some AI-specific regulations have already been implemented, they mainly rely on broader frameworks like data protection laws (e.g., the European Union's GDPR and the California Consumer Privacy Act (CCPA) in the United States) [44]. These regulations primarily focus on data privacy and fail to address the ethical challenges associated with AI decision-making comprehensively. To address this issue, various bias mitigation algorithms and open-source packages have been developed [11, 40].

The integration of XAI in various domains has significant societal impacts since XAI plays a crucial role in detecting and mitigating

biases, ensuring fairness, and facilitating accountability [26, 31, 35, 36, 41, 42, 49]. However, research indicates limited engagement with ethical theories in XAI frameworks, highlighting the need for more comprehensive ethical integration [33].

Hence, ethical considerations in AI include the following distinct yet related key insights [26, 31, 34, 36, 46, 49]:

- **Transparency and Trust:** Enhancing trust by clarifying and understanding the decision-making processes. Transparent systems ensure that users can comprehend and validate AI outcomes, avoiding arbitrary or opaque decisions.
- **Fairness and Bias Mitigation:** Ethical AI development involves addressing biases that could lead to discriminatory outcomes. Identifying and correcting errors, biases, or undesirable consequences is necessary. Mitigating bias requires diverse representation in AI design, regular audits, and adherence to ethical guidelines to ensure fairness across different demographic groups.
- **Data Privacy and Usage:** Safeguarding user privacy is a core principle of responsible AI. Anonymization, encryption, and informed consent protect users' personal data while complying with regulations like GDPR. Ethical data practices build user confidence and ensure respect for individual rights.
- **Accountability and Regulation:** AI documentation and explanations are needed to meet legal and ethical standards in regulated industries. Providing explanations for AI decisions makes it possible to understand, challenge, and hold AI systems and developers accountable for their actions. Clear accountability frameworks ensure that responsibility

is shared among developers, organizations, and users interacting with the system.

- **Collaboration and Informed Decision-Making:** There is a need to improve the collaboration between humans and AI by providing understandable insights. This empowers users to make informed decisions, understand system strengths and limitations, and enhance the efficiency of the decision-making process.
- **Human-Centric and Socially Responsible AI:** AI systems should prioritize human dignity, well-being, and societal values. Involving stakeholders throughout the systems' development process ensures that technologies are aligned with ethical standards, promote social justice, and minimize harm. Additionally, there is a need to address broader societal impacts such as inequality and job displacement.

Figure 1 presents the ethical considerations regarding AI and its main purpose.

3 XAI4RE – Theoretical Framework for Using XAI for Responsible and Ethical AI

XAI4RE suggests a theoretical framework for using XAI effectively through the AI lifecycle (e.g., design, development, and implementation) to address AI's key ethical considerations concerning the various stakeholders. The framework was created based on state-of-the-art XAI frameworks such as [1, 25, 31, 33, 35, 37, 45], relevant regulations such as [15, 44], and ethical frameworks such as [19, 26, 27, 36, 46, 49]. The following subsections refer to the AI lifecycle stages. Each sub-section describes the aim of the stage, the insights that can be drawn using XAI methods for the various stakeholders, and the ethical considerations to be addressed.



Figure 1: Ethical Considerations for AI

3.1 Stage 1: Problem Definition and Feasibility

This stage aims to identify the problem to be solved, define objectives, and assess the feasibility of using AI. Using XAI at this stage may help provide interpretable simulations of AI outcomes, assess feasibility, and visualize the potential impact of the design decisions on the fairness and ethics of the system. For example, developers may use XAI tools to predict system behavior and identify risks early. Regulators may review XAI-generated simulations to ensure compliance with ethical standards. Users can use XAI visualizations to understand the system's objectives and provide their input and/or concerns.

Hence, using XAI methods at this stage enables us to clarify the purpose and potential outcomes of the system (regarding the ethical consideration of Transparency and Trust), to identify risks and biases (regarding the ethical consideration of Fairness and Bias Mitigation), and to align AI objectives with societal values by engaging various stakeholders (regarding the ethical consideration of creating Human-Centric AI).

3.2 Stage 2: Data Collection and Preparation

This stage aims to gather and preprocess data to be used for training and validating the system. Using various XAI methods at this stage enables understanding data distributions, detecting under-represented groups or biased patterns, and generating reports on data quality, relevance, and fairness for regulatory audits. For example, developers, regulators, and users can use XAI methods at this stage to ensure the datasets are representative and free of bias, to validate data compliance with legal and ethical requirements, to gain insights on data relevance, and to flag under-discovered biases based on their domain expertise.

When referring to the ethical considerations of Data Privacy and Usage, Fairness and Bias Mitigation, and Accountability, the usage of XAI methods at this stage enables ensuring compliance with data protection regulations through transparency, detecting and addressing biases in datasets and documenting data collection processes for traceability, respectively.

3.3 Stage 3: Model Development and Training

The main aim of this stage is to develop and train the AI model using the collected data. At this stage, XAI methods such as SHAP, LIME, and counterfactual explanations can be used for providing feature importance metrics, identifying influential variables, creating interpretable dashboards to highlight biases during training, and comparing outputs with expected domain-specific norms. Developers may use XAI tools to refine the model and address unintended biases. Regulators may audit the training processes using XAI and verify alignment with domain ethical guidelines. Additionally, using XAI at this stage may create understandable explanations for users regarding how the system works, so users will be able to collaborate with the developers and evaluate early model outputs, practicality, and fairness.

Ensuring that the logic of the model is interpretable for developers and the other stakeholders, the ability to monitor biases during training, and the possibility of collaborative development using interpretable models aligning with the ethical considerations of

Transparency and Trust, Fairness and Bias Mitigation, and Collaboration and Informed Decision-Making, respectively.

3.4 Stage 4: Model Evaluation and Validation

This stage aims to test the model's performance, reliability, and alignment with objectives. Counterfactual explanations may be used to test the model's behavior under hypothetical conditions. Moreover, various XAI methods may generate transparent reports on how the input features influence model decisions. This will enable developers to validate model outputs and to identify weaknesses in the model. Detailed XAI reports, produced at this stage, may help regulators to evaluate the system's compliance with fairness standards and enable users to provide feedback on the evaluation results and to ensure alignment with real-world needs.

Providing interpretable outputs and comprehensive evaluation of that system, testing the model across diverse groups for unbiased performance, and documenting the evaluation results using detailed explanations can be related to the ethical considerations of Transparency and Trust, Fairness and Bias Mitigation, and Accountability.

3.5 Stage 5: Deployment

This stage deals with the integration of the AI system for real-world use. At this stage, XAI can be used to provide user-friendly explanations for the decision-makers and for developing interpretable dashboards to track the system's performance. Developers can use XAI at this stage to ensure that the deployed system is interpretable and user-friendly. Regulators will be able to monitor the deployment for compliance with relevant legal and ethical standards. Additionally, both decision-makers and end-users may leverage XAI explanations to build their trust and confidence in the system.

Using XAI at this stage enables clear communication regarding how the AI outputs are generated and their limitations. This understanding may help train users to interpret and apply AI outputs effectively, as well as to monitor the early deployment of the system and to ensure its alignment with societal values. Those benefits can be related to the ethical considerations of Transparency and Trust, Collaboration and Informed Decision-Making, and Human-Centric AI, respectively.

3.6 Stage 6: Monitoring and Maintenance

This stage aims to monitor the model's performance and maintain reliability post-deployment. XAI debugging tools that are used at this stage may help monitor the system's performance by using interpretable metrics and visualizations, and trace the causes of unexpected outcomes. For example, developers can use XAI to identify and resolve issues during system updates, regulators may conduct periodic audits using XAI tools to ensure the system's ongoing compliance. Users can report anomalies and suggest improvements based on XAI-driven insights.

At this stage, using various XAI methods may address the ethical considerations of Fairness and Bias Mitigation, Accountability, and Human-Centric AI by enabling continuous monitoring for emerging biases and unintended consequences of the system, ensuring transparency in updates, and retraining and adjusting the system to reflect the changing societal needs.

3.7 Stage 7: Retirement or Repurposing

This stage refers to the “phase out” or repurposing of models that are no longer effective or relevant. XAI can be used to generate final reports explaining the rationale for decommissioning. Furthermore, XAI analyses from the previous stages may be used to assess the potential reuse of the system in other contexts. Developers can manage the decommissioning with XAI tools to ensure responsible system retirement. Regulators can oversee the decommissioning process and validate its compliance using XAI reports. Users will be able to provide their insights regarding the system’s impact and evaluate the rationale for its retirement.

The relevant ethical considerations for this stage are Accountability, which is related to the ability to create proper documentation of decommissioning decisions. Transparency and Trust, which are relevant for communicating the reasons for retirement and the

potential impacts, and Human-Centric AI, in which XAI can be used for evaluating the opportunities for repurposing.

A visualization of XAI purposes as presented in the XAI4RE theoretical framework and a summarization of the suggested general guidelines for each stage are presented in Figure 2 and Figure 3, respectively.

4 XAI4RE Implementation

This theoretical framework will include the following steps: (1) Identify the context of the system, its objectives, and the relevant stakeholders. (2) Understand stakeholders’ expectations and define explainability goals for each AI lifecycle stage with the help of the suggested guidelines. (3) Integrate various XAI methods throughout the AI lifecycle stages. (4) Create tailored explanations for diverse stakeholder roles and expertise levels. (5) Test and validate XAI clarity and utility using stakeholders’ feedback

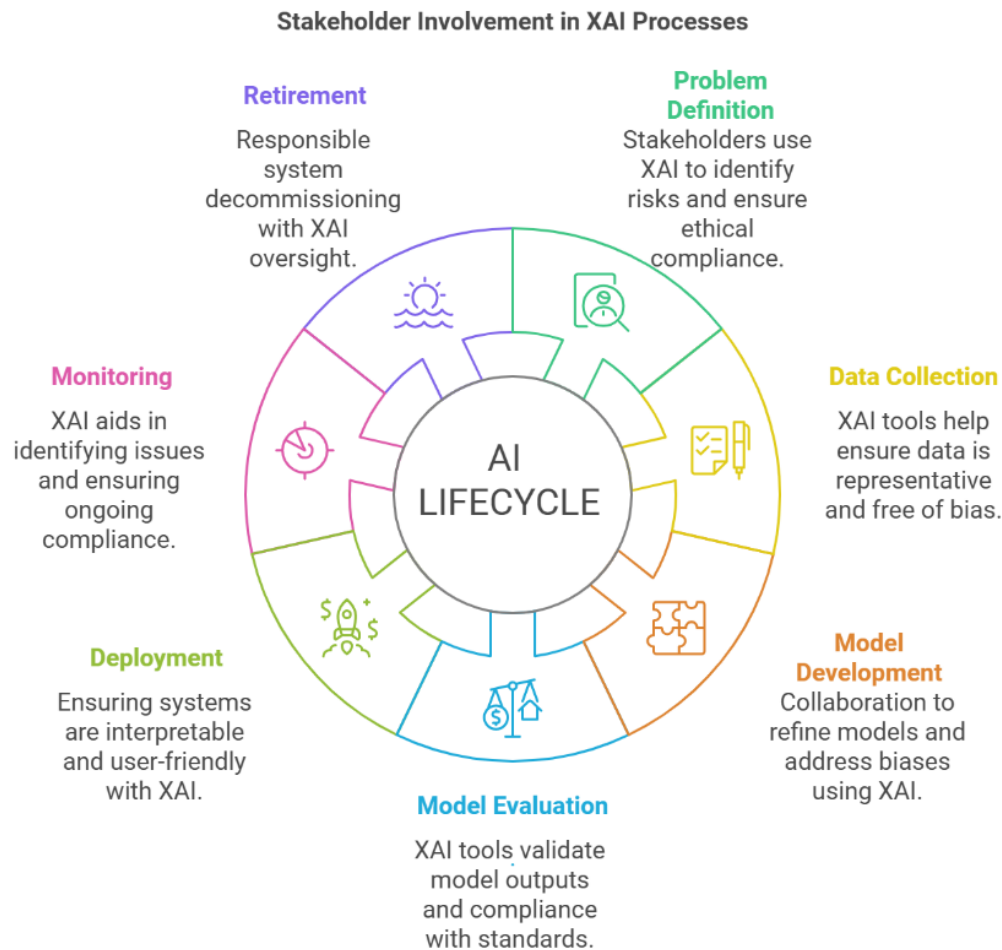


Figure 2: XAI objectives at each stage of the AI lifecycle.

XAI4RE GUIDELINES SUMMARIZATION	Problem Definition and Feasibility	Data Collection and Preparation	Model Development and Training	Model Evaluation and Validation	Deployment	Monitoring and Maintenance	Retirement or Repurposing
Transparency and Trust	X		X	X	X		X
Fairness and Bias Mitigation	X	X	X	X		X	
Data Privacy and Usage		X					
Accountability and Regulation		X		X		X	X
Collaboration and Informed Decision-Making			X		X		
Human-Centric and Socially Responsible AI	X				X	X	X
XAI Applications	Simulate and visualize outcomes, assess AI fairness and ethics	Analyze data distributions, expose biases, highlight influential features	Use feature importance tools, detect training biases, provide interpretable visualizations	Generate transparent metrics, use counterfactuals, explain feature impacts	Provide user-friendly explanations, interpretable dashboards, communicate limitations	Monitor root causes, trace for biases and unexpected outcomes, document changes	Explain decommissioning, analyze reuse opportunities, document lessons
XAI Purpose	Clarify AI goals, identify risks, build trust	Ensure data quality, fairness, and transparency.	Monitor fairness, guide development, and align outputs with expectations.	Validate robustness, fairness, and adherence to ethical standards.	Enhance trust, empower decision-makers, and promote responsible system usage.	Maintain accountability, fairness, and system reliability.	Ensure ethical end-of-life management, foster transparency, and guide future AI.
XAI for Developers	Define feasibility, identify risks	Handle data preprocessing, labeling	Train models, monitor for bias	Test performance, document outcomes	Oversee integration, monitor performance	Audit models, implement updates	Manage decommissioning, assess reuse potential
XAI for Regulators	Set ethical guidelines, oversee compliance	Audit practices, ensure legal compliance	Set standards for fairness and explainability	Enforce compliance, review metrics	Audit deployment, ensure standards are met	Monitor compliance, conduct reviews	Ensure data and compliance standards are upheld
XAI for Users	Provide context, needs, and use case insights	Validate data relevance, highlight biases	Offer feedback on usability	Participate in validation and feedback	Validate usability, report issues	Report errors, contribute to refinements	Evaluate system impact, provide feedback

Figure 3: Summarization of XAI4RE Suggested Guidelines. The table provides general guidelines on insights that stakeholders can derive from XAI, along with the ethical considerations relevant to each stage of the AI lifecycle.

loop and/or benchmarking. (6) Use XAI’s explanations for ethical assessment to ensure systems’ compliance with relevant legal and ethical standards and with societal values. And (7) Record all XAI integration and compliance aspects for continuous monitoring and future feedback loops. Steps 3,4,5 can be performed iteratively to assess the suitability of the explanations provided to the explanation objectives for each stage of the life cycle. In step 5, different XAI evaluation frameworks, such as [37], can be used to evaluate explanations. Moreover, step 5 also requires collaboration between the various stakeholders to ensure that the gained insights and conclusions are understandable and consistent.

The framework ensures that XAI is applied at every stage, from problem definition to retirement, promotes collaboration among developers, regulators, users, and domain experts, directly addresses transparency, trust, fairness, accountability, and human-centered AI, ensures continuous monitoring and adjustment to align with societal values, facilitates user understanding and fosters trust in AI systems. The suggested guidelines regarding the purpose of using XAI at each stage can be implemented into an evaluative checklist that researchers and professionals can systematically apply. For example, considering a real-world scenario such as a healthcare diagnostic system, we may apply the XAI4RE framework as follows: Stage 1 (Problem Definition): Use XAI to simulate patient diagnosis outcomes and identify ethical risks. Stage 2 (Data Collection): Monitor data quality and bias using XAI reports. Stage 3 (Model Development): Apply SHAP to identify key symptoms influencing diagnoses. Stage 4 (Model Evaluation): Generate comprehensive validation reports with transparent explanations. Stage 5 (Deployment): Create patient-friendly dashboards explaining diagnostic decisions. Stage 6 (Monitoring): Continuously assess diagnostic accuracy and explanation fidelity. Stage 7 (Retirement): Generate a final report justifying model decommissioning. At each stage, the suitability of the explanations for the various stakeholders should

be assessed. Then, use the explanations themselves and stakeholder feedback to assess compliance with the relevant ethical considerations for that stage and document what is being done for monitoring purposes.

Hence, the suggested theoretical framework and guidelines ensure that XAI will support AI ethical considerations effectively, engage all stakeholders, and promote responsible, ethical, and human-centered AI development and deployment while promoting trust, accountability, and collaboration among the various stakeholders.

5 Conclusions

As AI systems increasingly influence critical decisions, ensuring transparency, fairness, and accountability becomes essential to fostering user trust and mitigating potential harm. The rapid advancement of AI technologies has outpaced current regulatory efforts, leaving gaps in accountability, bias mitigation, and transparency. To address these challenges, there is a pressing need for regulations that specifically tackle the ethical aspects of AI, promoting responsible development and mitigating potential risks [36]. These challenges underscore the importance of using various XAI methods and adopting ethical frameworks to guide AI development and deployment. Hence, integrating XAI into the AI lifecycle is not just a technical imperative but an ethical necessity. XAI provides actionable insights that enable various stakeholders to comprehend and validate AI decisions, ensuring accountability and trustworthiness, supporting bias detection and mitigation, promoting equitable outcomes across diverse groups, and ensuring responsible data use while protecting user privacy. Effective XAI implementation requires balancing technical and ethical dimensions [26] in order to bridge the gap between expert-focused explanations (i.e. explanation that focuses on debugging, reliability, and validation of the system, how the system works?) and society-oriented explanations

(i.e. explanation that build trust, enhance transparency and fairness, comply with regulatory and policy changes, and act as external and social validation. Why was the decision made?) [14]. Recently, the importance of considering both technical and ethical dimensions of XAI has arisen significantly [16, 24, 26].

Despite its benefits, a fundamental problem arises from using XAI. Unintended negative effects of XAI have been identified. XAI systems can be manipulated or misused and potentially mislead users to trust the system or manipulate them to support predetermined conclusions [9, 20]. Studies have shown that deceptive explanations can be generated to manipulate user trust in AI systems [23, 37]. Non-technical users may develop an illusion of understanding when interpreting XAI explanations [7, 42]. Common evaluation methods for XAI systems, such as proxy tasks and subjective measures, can be misleading and may not accurately predict real-world performance [4]. Explanations can become manipulation tools in persuasive technologies [5] and more.

Various XAI frameworks and ethical frameworks have been created lately. For example, Arrieta et al. [1] presented the diverse purposes of using XAI techniques and argued that stakeholders are a key aspect that needs to be considered when using XAI and that using XAI should be one of the ways to establish responsible AI. They also emphasize that the main challenge remains in realizing the potential of XAI for responsible AI. Naiseh et al. [33] proposed an XAI framework that enables collaboration between different stakeholders from different disciplines. Their method involves identifying representative stakeholders who can provide insights from the explanations, identifying potential risks in trust calibration while iteratively evaluating both technical and human perspectives, and mitigating the potential risks. Furthermore, they argued that their framework can help users identify trust calibration challenges and foster empathy towards the users of XAI interfaces. Furthermore, Yadav [49] proposed a methodology for investigating the ethical dimensions of XAI with the aim of identifying key ethical issues, understanding stakeholder perspectives, developing ethical frameworks, and producing policy recommendations. He also argued that there is a need to produce recommendations for policymakers on regulating XAI applications and to promote wider adoption of ethical AI practices to increase users' trust and the effectiveness of AI.

This paper suggests an XAI4RE theoretical framework that explores how to integrate XAI principles at the various stages of the AI lifecycle with respect to diverse stakeholders' needs. It demonstrates how various XAI methods can serve as a vital tool for making AI systems comprehensible and ethically aligned with societal values. The proposed framework illustrates how XAI can help address various ethical considerations at each stage of the AI lifecycle while balancing the diverse needs of developers, regulators, and users. Developers can use XAI4RE to incorporate ethical considerations throughout AI development; Regulators could leverage it to evaluate compliance with AI ethics policies; Users (decision-makers and end-users) could benefit from improved transparency and control over AI decisions. By involving various stakeholders throughout the AI lifecycle, the XAI4RE theoretical framework may align AI systems with societal values, minimizing harm and enhancing stakeholders' collaboration. XAI4RE theoretical framework and its suggested guidelines align with both XAI frameworks

(e.g., [1, 28, 33, 39, 45, 49]) and ethical frameworks (e.g., [11, 16, 21]). However, as far as we know, none of the above frameworks refer to all aspects related to ethical considerations for each stage of the AI lifecycle with respect to different stakeholders.

XAI also faces challenges, including the risk of manipulation or incomplete explanations (e.g. misleading users to trust the system or manipulating them to support predetermined conclusions [9, 20, 23, 37, 38], non-technical users may develop an illusion of understanding when interpreting XAI explanations [7, 42]; XAI explanations may not accurately predict real-world performance [4]; the explanations can become manipulation tools in persuasive technologies [5] and more). However, the XAI4RE framework can be used to solve this challenge by using XAI throughout the AI lifecycle with full transparency between the various stakeholders. Addressing this challenge requires continued interdisciplinary collaboration to refine XAI methodologies, integrate ethical theories, and adapt frameworks that address stakeholders' needs. Hence, XAI4RE represents a theoretical framework that can deal with these challenges and can be practically implemented to create responsible, ethical, and human-centered AI, ensuring that technological innovation remains transparent, ethical, and aligned with human values.

6 Limitations and Future Work

This paper underscores the importance of integrating ethical considerations in AI systems and the ways that XAI can help to do so with respect to different stakeholders and the different AI lifecycle stages. While XAI can significantly enhance user understanding and trust, it also poses a major problem: the possibility of using XAI to mislead users, intentionally or unintentionally. However, the problem itself may also be the solution. This duality emphasizes the importance of involving XAI ethical considerations in developing and deploying AI systems. Hence, it is essential to develop human-in-the-loop evaluation methods, customize them for different stakeholders, ensure ethical oversight, and align them with legal and ethical standards. The XAI4RE framework can help deal with challenges like potential misuse of explanations, balancing interpretability with performance, and adapting to varied stakeholder needs. As a result, XAI4RE theoretical framework can effectively promote ethical and responsible AI systems.

In the future, we will focus on the empirical evaluation of the XAI4RE framework (i.e., developing a system with and without using the XAI4RE framework and comparing the alignment of both systems to ethical considerations relevant to the system domain). Future research should focus on advancing XAI techniques, fostering regulatory and technical cooperation, and conducting user-centric studies to improve real-world implementation. The theoretical framework presented in this paper serves as a foundation for navigating these challenges, promoting trust and accountability in AI systems while maintaining full transparency between the various stakeholders. Additionally, it is necessary to continue conducting multidisciplinary, human-centered research related to XAI's technical and social implications to ensure responsible and ethical AI systems.

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